

Fisher's 3 principles of experimental design

Presentation Outline

- 1 R.A. Fisher
- 2 Fisher's 3 principles of experimental design
 - Randomization
 - Replication
 - Local control of variability



R.A. Fisher, 1890-1962



- 1912: Graduated in math, Cambridge
- 1919-1933: Hired to “make sense of data” at Rothamsted Experimental Station
- 1931, 1935: Visited ISU
- 1933 - 1957: Professor of Genetics, UCL, Camb.
- Foundational contributions in both statistics and genetics/evolutionary biology
- Experimental design, ANOVA, F distribution
- *Statistical Methods for Research Workers*, 1925, 14'th ed. 1970
- *The Design of Experiments*, 1935, 9'th ed. 1971

Fisher's 3 principles

- Randomization
- Replication
- “Local control of variability” - now known as blocking

Randomization

- Randomly assign experimental units to different treatments
- The experimental unit is what is randomized to a treatment
 - schools, cows, plants, pens, plots, beakers
- We'll talk a lot more about experimental units when we discuss the independence assumption

The randomization controversy

- Very controversial when proposed in the 1920's / 1930's
- Context: field experiments in agronomy
- Agronomists preferred systematic designs: A, B, A, B, A, B ...
 - Or a checkerboard arrangement
- Claimed it gave more precise estimates of the treatment difference(s)
- Fisher argued for randomization, e.g., A, B, B, A, A, B ...
- Resolved by analyzing uniformity trial data
 - Variability between plots in a field without treatments
 - Supported Fisher

Replication

- Randomly assigning a treatment to multiple experimental units
- Randomization and replication required for valid estimates of the experimental error
 - Only based on how the data were collected (design-based inference)
 - Other methods require assuming that a model is correct (model-based inference)

Local control of variability

- Optional feature of a study design
 - Group experimental units into blocks of similar pre-treatment characteristics
 - Randomly assign treatments within each block
- Usually gives more precise estimates of treatment differences
- Pairing is blocking with 2 treatments and blocks of size 2
- Will talk a lot more about pairing in module 2 (week 3)
and blocking in general in module 7 (week 14/15)

Important Points:

Important Points:

- Randomization
 - Randomly assigning treatments to expt. units
 - Allows causal claims based on the data
- Replication
 - Provides valid measures of variability
- Blocking
 - When done well, reduces unwanted variability
 - Pairing is 2 treatments in blocks of size 2